

OPTICROP: SMART AGRICULTURAL PRODUCTION OPTIMIZATION ENGINE

PROJECT USER & SYSTEM MANUAL

1. Source Repository & Cloning

The official source repository is hosted on GitHub. Use the link below to view or clone the code assets:

- GitHub Link: <https://github.com/goutham-05-raj/OptiCrop-Smart-Agricultural-Production-Optimization-Engine>
- Clone command: `git clone https://github.com/goutham-05-raj/OptiCrop-Smart-Agricultural-Production-Optimization-Engine`

2. Setup Instructions

Follow these steps to run the OptiCrop engine on a local machine:

- Prerequisites: Install Python 3.10+ and git version control.
- Setup Environment: Run `'python -m venv .venv'` and activate it.
- Install Packages: Run `'pip install -r requirements.txt'`.
- Data & Models: Run `'python generate_datasets.py'` and `'python train_models.py'` to generate model files.
- Run Server: Run `'python app.py'` and open `'http://localhost:5000'` in a web browser.

3. Operating Instructions

Enter NPK and environmental values in the form, then click 'Execute Production Optimization'. Review the suggested crop and fertilizer recommendations.